HW7

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NOTE: I have a Mac so all SAS is run on SAS Studio online. So, I will copy any code I use into a code block here, but all outputs I will need to provide screenshots/regenerate them in R.

1

Y and e are written with subscripts of block,treatment, ie y_{12} is block 1 treatment 2.

$$Y = \begin{pmatrix} y_{11} \\ y_{12} \\ y_{21} \\ y_{22} \\ y_{31} \\ y_{32} \end{pmatrix}$$

X is n by num_treatments. Which 'treatment' is each response from?

$$X = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \\ 1 & 0 \\ 0 & 1 \end{pmatrix}$$

$$\beta = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}$$

Z is n by num_blocks. Which 'block' is each response from?

$$Z = \begin{pmatrix} 1 & 0 & 0 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 1 \end{pmatrix}$$

$$u = \begin{pmatrix} u_1 \\ u_2 \\ u_3 \end{pmatrix}$$

$$e = \begin{pmatrix} e_{11} \\ e_{12} \\ e_{21} \\ e_{22} \\ e_{31} \\ e_{32} \end{pmatrix}$$

2

```
set.seed(1337)
treat_means <- c(100, 110)
sig2_e <- 16
sig2_b <- 12
n_treat <- 2
n_block <- 10
n_samp <- 5
u <- rnorm(n_block, 0, sqrt(sig2_b))

gen_data <- function() {
    treatment <- factor(paste0("Trt_", 1:n_treat))
    block <- factor(paste0("Blk_", 1:n_block))
    reps <- 1:n_samp

    design_matrix <- expand.grid(Rep = reps, Treatment = treatment, Block = block)

    y <- unlist(lapply(1:n_treat, \(i) lapply(1:n_block, \(j) lapply(1:n_samp, \(k) treat_mea
    dat <- as.data.frame(cbind(y, design_matrix$Block, design_matrix$Treatment))
    colnames(dat) <- c("y", "Block", "Treatment")
    dat$Block <- as.factor(dat$Block)
    dat$Treatment <- as.factor(dat$Treatment)
```

```
    dat
}

dt <- gen_data()
write.csv(dt, 'q2_gen_data.csv', row.names = FALSE)
```

Not going to show the generated data, but I export it to CSV to get into SAS Studio.

3

```
PROC IMPORT DATAFILE='/home/u49685591/STAT537/HW7/q2_gen_data.csv'
    DBMS=CSV
    OUT=WORK.HW7
    REPLACE;
    GETNAMES=YES;
RUN;

proc mixed data=HW7 method = REML;
class Block Treatment;
model y = Treatment / noint s ddfm=satterth;
random Block;
run;
```

```
sas_sigma2_b <- 30.4196
sas_sigma2_e <- 27.3545
sas_trt1_mean <- 107.17
sas_trt2_mean <- 108.31
```

The estimated output from SAS indicates a σ_b^2 of 30.4196 and a σ_e^2 of 27.3545, which are nowhere near the actual values provided. It also estimated that $\beta_{Trt1} = 107.17$ and $\beta_{Trt2} = 108.31$, which does show Treatment 1 being lower than Treatment 2, but not by much (the typical interpretation of Treatment 2 as a difference of Treatment 1 would be $\beta_{Trt2} = 1.14$).

4

```
sas_fixed_f_stat <- 1751.09
```

The F Statistic provided by the output of SAS for the test of whether there is a significant treatment effect is 1751.09. This gave a P-value WELL under .05 which would indicate that yes, treatment does have a significant effect.

5

```
suppressPackageStartupMessages(library('lme4'))
mod <- lmer(y ~ Treatment + (1|Block), data = dt)
summary(mod)
```

```
Linear mixed model fit by REML ['lmerMod']
Formula: y ~ Treatment + (1 | Block)
   Data: dt

REML criterion at convergence: 632.7

Scaled residuals:
     Min       1Q   Median       3Q      Max
-2.67991 -0.51338  0.07664  0.65787  2.05296

Random effects:
 Groups   Name        Variance Std.Dev.
 Block    (Intercept) 30.42    5.515
 Residual             27.35    5.230
Number of obs: 100, groups:  Block, 10

Fixed effects:
            Estimate Std. Error t value
(Intercept)  107.165      1.894  56.567
Treatment2     1.146      1.046   1.096

Correlation of Fixed Effects:
           (Intr)
Treatment2 -0.276
```

```
# Outputs from Summary
r_lmer_sigma2_b <- 30.42
r_lmer_sigma2_e <- 27.35
r_lmer_trt1_mean <- 107.165 # Intercept of the fixed effects
r_lmer_trt2_mean <- 105.704 + 1.146
```

We got the exact same values (within rounding) as the output from SAS.

The estimated output from R indicates a σ_b^2 of 30.42 and a σ_e^2 of 27.35, which, again, are nowhere near the actual values provided. It also estimated that β_{Trt1} = 107.165 and β_{Trt2} = 106.85, which does show Treatment 1 being lower than Treatment 2, but not by much (the typical interpretation of Treatment 2 as a difference of Treatment 1 would be β_{Trt2} = -0.315).

6

```
mod_ml <- lmer(y ~ Treatment + (1|Block), data = dt, REML = FALSE)
summary(mod_ml)
```

```
Linear mixed model fit by maximum likelihood  ['lmerMod']
Formula: y ~ Treatment + (1 | Block)
   Data: dt

      AIC       BIC    logLik -2*log(L)  df.resid
    645.6     656.0    -318.8     637.6        96

Scaled residuals:
     Min       1Q   Median       3Q      Max
-2.70364 -0.52539  0.07862  0.66560  2.07168

Random effects:
 Groups   Name        Variance Std.Dev.
 Block    (Intercept) 27.13    5.209
 Residual             27.05    5.201
Number of obs: 100, groups:  Block, 10

Fixed effects:
            Estimate Std. Error t value
(Intercept)  107.165      1.804  59.404
Treatment2     1.146      1.040   1.102

Correlation of Fixed Effects:
           (Intr)
Treatment2 -0.288
```

```r
# Outputs from Summary
r_ml_sigma2_b <- 27.13
r_ml_sigma2_e <- 27.05
```

The β estimates actually remained exactly the same. However, the estimates of the variance changed to σ_b^2 of 27.13 and a σ_e^2 of 27.05 which is still not close to the true values, though technically closer.

7

```r
suppressPackageStartupMessages(library('lmerTest'))

# I'm using this package I found online to test the likliehood since ANOVA doesn't work with
# Test using the REML method
ranova(mod)
```

```
ANOVA-like table for random-effects: Single term deletions

Model:
y ~ Treatment + (1 | Block)
             npar  logLik    AIC    LRT Df Pr(>Chisq)
<none>          4 -316.33 640.66
(1 | Block)     3 -339.59 685.17 46.511  1   9.11e-12 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```r
# Test using the ML method
ranova(mod_ml)
```

```
ANOVA-like table for random-effects: Single term deletions

Model:
y ~ Treatment + (1 | Block)
             npar  logLik    AIC    LRT Df Pr(>Chisq)
<none>          4 -318.78 645.57
(1 | Block)     3 -341.51 689.03 45.463  1  1.556e-11 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Both tests indicate that, with p-values MUCH less than .05, we reject $H_0 : \sigma_b^2 = 0$. I didn't do anything specific to divide by 2, but I'm assuming the ranova() function already does that.

8

```
mod_red <- lmer(y ~ (1 | Block), data = dt, REML = FALSE)
anova(mod_ml, mod_red)
```

```
Data: dt
Models:
mod_red: y ~ (1 | Block)
mod_ml: y ~ Treatment + (1 | Block)
        npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
mod_red    3 644.77 652.59 -319.39    638.77
mod_ml     4 645.57 655.99 -318.78    637.57 1.2057  1     0.2722
```

Comparing the reduced to the full model, we get a χ^2 statistic of ~1.21 with degrees of freedom of 1, and p-value of ~.27 which indicates that we accept the $H_0 : \beta_{Trt1} = \beta_{Trt2}$. The F-test said this was definitely significant, where as this test says it is not significant.

9

```
remove_rows <- sample(1:100, 20, replace = FALSE)
dt_red <- dt[-remove_rows,]
write.csv(dt_red, 'q9_gen_data.csv', row.names = FALSE)
```

```
PROC IMPORT DATAFILE='/home/u49685591/STAT537/HW7/q9_gen_data.csv'
    DBMS=CSV
    OUT=WORK.HW7
    REPLACE;
    GETNAMES=YES;
RUN;

proc mixed data=HW7 method = REML;
class Block Treatment;
model y = Treatment / noint s ddfm=satterth;
random Block;
run;
```

In the final Fixed Effects test, the DF changed from 15 in the first analysis to 14.6 in this one.

10

The degrees of freedom changed a bit based on how the data were removed. In this case I got a DF of 11 for Treatment 2 and 10.5 for Treatment 1, compared to 10.5 for both in the first analysis.

11

```
PROC IMPORT DATAFILE='/home/u49685591/STAT537/HW7/q9_gen_data.csv'
    DBMS=CSV
    OUT=WORK.HW7
    REPLACE;
    GETNAMES=YES;
RUN;

proc mixed data=HW7 method = REML;
class Block Treatment;
model y = Treatment / noint s ddfm=KenwardRoger;
random Block;
run;
```

I noticed that the DF for the test of fixed effects jumped up to 20.4, where the previous test used a DF of 14.6. This also impacted the F statistic, though the p-value is still VERY small.

12

I perform this back on the original dataset, not the one with 20 removed rows.

```
mod_int <- lmer(y ~ Treatment + (1 | Block) + (1 | Treatment:Block), data = dt)
anova(mod, mod_int)
```

```
refitting model(s) with ML (instead of REML)
```

```
Data: dt
Models:
mod: y ~ Treatment + (1 | Block)
mod_int: y ~ Treatment + (1 | Block) + (1 | Treatment:Block)
        npar    AIC    BIC  logLik -2*log(L)  Chisq Df Pr(>Chisq)
mod        4 645.57 655.99 -318.78    637.57
mod_int    5 623.73 636.76 -306.86    613.73 23.836  1  1.049e-06 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

Even though we know the interaction doesn't exist, we do indeed reject the $H_0 : \sigma^2_{tb} = 0$ and we conclude that there is a significant interaction.